



This assessment grades DentOS against industry standards for dental practice-management software, and against its real target market: small-to-mid private clinics in Pakistan and the wider region. It is written through two lenses at once — a practising-dentist view of the clinical record, and a project-delivery view of what it takes to put software in front of paying clinics.

One caveat: it is based on the system as built and discussed, not a line-by-line code audit. Anything already implemented should be read as “confirm in place” rather than “missing.”

PRIORITY SNAPSHOT

AREA	ITEM	PRIORITY
Platform	Windows support	Blocker
Data	Migration safety + automated tests	Blocker
Billing	Partial payments, ledger, day-sheet	Blocker
Data	Proven backup / restore + disaster recovery	Blocker
Sync	Conflict handling (last-write-wins risk)	Blocker
Clinical	Progress / SOAP clinical notes	Core
Clinical	Periodontal charting (pocket depths, BOP)	Core
Clinical	Imaging — store / view X-rays + photos	Core
Clinical	Medical history, medications, ASA flags	Core
Clinical	Prescriptions and signed consent forms	Core
Clinical	Dental lab case tracking	Core
Growth	Appointment reminders (WhatsApp norm)	High
Growth	Recall / recare automation	High
Growth	Data import / onboarding from old records	High
Growth	Owner reports: A/R ageing, production	High
Eng	Automated tests (sync, billing, schema)	High
Eng	Crash / error reporting + logging	Medium
Eng	Role-based access enforcement audit	Medium
Eng	In-app auto-update mechanism	Medium

Blocker fix before any clinic relies on it | **Core** needed to be a true clinical record | **High** competitive & adoption | **Medium** quality & operations

01 Where DentOS is genuinely strong

- **Local-first, encrypted core.** An encrypted SQLCipher database that runs fully offline is a real advantage where power and connectivity are unreliable — the clinic keeps working when the internet does not.
- **Cloud sync and multi-branch.** Supabase sync with row-level security and per-branch scoping gives a credible path from a single practice to a multi-branch group.
- **Monetisable licensing.** The RSA-signed license and automated provisioning make this something you can actually sell and control, not just an internal tool.
- **Modern interface.** A clean, current UI is a genuine differentiator against both the dated global incumbents and the crude local alternatives.
- **Charting foundation.** An FDI odontogram plus a 3D model viewer is a strong, marketable starting point for clinical charting.

02 Deployment blockers

These cause data loss, money errors, or “we can’t even install it.” They come before everything else.

- **Windows support.** The overwhelming majority of clinics in the target market run Windows. A macOS-only build limits the addressable market to a fraction of it — the single highest-impact item for real-world reach.
- **Migration safety.** The schema-version drift already flagged is exactly the failure that corrupts live patient data on update day. A harness that upgrades old databases through every version is essential before any clinic depends on it.
- **Payments depth.** Real billing needs partial payments, a per-patient ledger with running balance, a daily cash close (day-sheet), printable receipts, and local methods such as cash, EasyPaiza and JazzCash. Money must reconcile to the rupee and be auditable.
- **Proven backup and recovery.** Backups are only real once a restore has been shown to rebuild a working clinic. Add tested restore and an offsite disaster-recovery path.
- **Sync conflict handling.** Last-write-wins silently drops the losing edit. With two front-desk users touching the same record, that is quiet data loss — detect and surface conflicts, especially on financial rows.

03 Clinical gaps

Today it is a strong front-office and restorative-charting system. To be a clinical system-of-record it needs:

- **Progress / SOAP notes.** Structured, templated per-visit notes are a medico-legal necessity and the backbone of a clinical record — if it is not documented, it did not happen.

- **Periodontal charting.** Six-point pocket depths, bleeding on probing, recession and mobility, tracked over time. The current odontogram covers restorative status only; perio is standard of care and absent.
- **Medical history.** A medication list, systemic conditions and ASA status with contraindication warnings should extend the existing allergy field.
- **Imaging.** Radiographs and intraoral photos need to be stored and viewed against the patient and tooth — today X-rays are billed but not kept.
- **Prescriptions and consent.** Printable prescriptions and digitally stored, signed consent forms.
- **Lab case tracking.** Crowns, dentures and implants go out to a dental lab with due dates and come back — a daily workflow your catalogue already implies.

04 Operational and growth

- **Appointment reminders.** No-shows are the largest avoidable revenue leak, and WhatsApp reminders are the market norm. Even a manual, pre-filled WhatsApp hand-off would materially cut no-shows.
- **Recall automation.** The recall-due status already exists; closing the loop with a worklist and reminders turns it into recovered revenue.
- **Data import and onboarding.** Clinics will not switch without importing existing patients from spreadsheets or older software; an importer removes the biggest barrier to adoption.
- **Owner financial reporting.** Accounts-receivable ageing, production versus collection, and per-provider productivity are the reports an owner buys the software to see.
- **Updates and localisation.** A safe in-app update path, Urdu labels for front-desk staff, and any required tax fields on invoices round out real-world fit.

05 Engineering and process

- **Automated tests.** Coverage for sync, migrations and billing maths is not optional for software holding medical and financial records.
- **Crash and error reporting.** Centralised error reporting and structured logs so failures in the field are visible.
- **Access-control audit.** Confirm roles gate clinical and financial data at the data layer, not merely in the UI.
- **Scale testing.** Validate performance at ten-thousand-plus patients and several years of history.
- **Record retention.** Make long-term retention (longer for minors) an explicit, documented rule; the soft-delete and audit log already support it.

06 Real-world verdict

The opportunity is real and specific: affordable, modern, offline-resilient, encrypted, multi-branch software for clinics the global vendors price out and the local vendors serve poorly. As it stands, DentOS is an excellent scheduling, charting and basic-billing product — a single-doctor clinic could pilot it today. It is not yet a full system-of-record, and the distance to that is gated by five things:

Windows support, payments and ledger depth, clinical and periodontal notes, appointment reminders, and migration safety. Close those, and a multi-chair or multi-branch practice can run on it.